Opposite exceptional ends after a later unshared cut

Human Compression project: bounded L044 continuation

We retain the standing L044 hypotheses: $G \le A_n$, $n \ge 12$, is a transitive string C-group with connected diagram and every maximal omission intransitive. The alternating rank bound at smaller degrees at least 12 is an explicit induction hypothesis. Assume $r > \lfloor (n-1)/2 \rfloor$ and an actual extendable six-point source Table 2(2) or (3) head at the first three labels. The accepted critical-head theorem gives $n = 2r \ge 18$.

Proposition 1 (What remains at a later unshared cut)**.** *Suppose $j \ge 5$ has a unique eligible crossing, with actual supported transitive sides $P = G_{<j}$ on C and $Q = G_{>j}$ on F, the original head contained in C. Then precisely the following alternatives remain unexcluded here:*

1. *$j = r - 1$, F is a singleton, and $P = \mathrm{Alt}(C) = A_{n-1}$ naturally;*
2. *$j = r - 3$, Q is the degree-five D_{10} marking, and $P = \mathrm{Alt}(C) = A_{n-5}$ naturally. The complete attachment theorem makes the actual last three marks a six-point Table 2 head.*

In particular an opposite actual six-point A_5 side is impossible. These are residual alternatives, not assertions that either exists.

Proof. The large-pure-prefix theorem identifies $P = \mathrm{Alt}(C)$; write $c = |C|$, $f = |F|$, and $d = \mathrm{rank}(Q)$. It also supplies the endpoint anchors and local intransitive omissions. Here $c \ge 12$: for $8 \le c < 12$ we have $j \ge 5 \ge (c-1)/2$. Apply the complete small primitive *prefix exclusion* (Theorem 3 of *Small primitive attachments through a decorated tail*), not the raw Table 2 list. Its hypotheses hold because P is a proper actual prefix, the ambient degree is at least 12, and the ambient marking is even, connected and has intransitive omissions. It excludes in particular the degree-ten rank-five and degree-eleven rank-six rows, leaving only the degree-five or degree-six exceptions, impossible on C. The small anchored imprimitive bound covers a nonprimitive side. Thus induction gives $j \le \lfloor (c-1)/2 \rfloor$.

Except for a singleton, the full dual of that proper-prefix exclusion and the proper primitive and anchored imprimitive bounds give $d \le \lfloor (f-2)/2 \rfloor$, unless Q is a degree-five D_{10}, a degree-six A_5 head, or a full natural alternating group of degree at least 12. The strict inequality case gives $r = j + d + 1 \le (n-1)/2$, contrary to the assumption. When both sides are large natural alternating groups, the complete faithful even one-point extension lemma gives an actual side marking of degree $c+1$ and rank $j+1$, or its counterpart on F. Its complete smaller-degree bound, added to the other side's bound, again gives $r \le \lfloor (n-1)/2 \rfloor$. This is the exact paid two-natural-side argument, including its projection-kernel proof.

For an opposite A_5 side, reverse the entire string. Its first two marks generate the degree-five dihedral group with trivial centralizer on that orbit. The crossing therefore fixes those five points and has the form $(v\ b)\delta$. Its actual opposite suffix is $P = \mathrm{Alt}(C)$. The accepted pure-head/natural-suffix obstruction applies and excludes it. The impossible reversed row 2 orientation is excluded by the literal head-extension lemma; row 3 has its stated reversing conjugation.

Only the singleton and the D_{10} alternative remain. Their ranks are respectively zero and two, so $j = r-1$ or $r-3$. In the latter case the complete six-choice dihedral attachment, applied to the globally reversed string, gives the stated actual last three marks. It does not assert that the next cut is unshared. □

Lemma 2 (The actual middle between two A_5 ends)**.** *Suppose the actual first and last three marks are extendable six-point Table 2 heads. Let their supports be D_L, D_R, and let v_L, v_R be their exit vertices. The supports are disjoint, and*

$$G_{\{3,\dots,r-4\}} = \mathrm{Alt}(Z), \qquad Z = \Omega \setminus ((D_L \setminus \{v_L\}) \cup (D_R \setminus \{v_R\})).$$

This is an actual natural alternating group of degree $n-10$ and rank $r-6 = (n-12)/2$. It has both endpoint anchors and all maximal omissions intransitive on Z.

Proof. Since $r \geq 9$, the two heads commute. A permutation centralizing either transitive six-point head preserves its unique nontrivial orbit and fixes that orbit pointwise, by its trivial centralizer. Thus the heads have disjoint supports. The complete critical-head theorem and its full dual give

$$G_{\geq 3} = \mathrm{Alt}(\Omega \setminus (D_L \setminus \{v_L\})), \qquad G_{\leq r-4} = \mathrm{Alt}(\Omega \setminus (D_R \setminus \{v_R\})).$$

Their natural permutation-group intersection is $\mathrm{Alt}(Z)$; full ambient IP identifies it with the displayed middle interval. No quotient IP is used. Every middle mark except its first fixes v_L, and every one except its last fixes v_R. If one of its maximal omissions were transitive on Z, adjoining the two full heads would make the corresponding ambient omission transitive on all of Ω, a contradiction. The rank count is literal. For even degree $n-10 \geq 12$ it is equality in, not a contradiction to, the smaller-degree alternating rank bound. □

Lemma 3 (A bounded exclusion within the two-ended alternative)**.** *The preceding two-ended configuration requires $n \geq 24$.*

Proof. Use the left critical-head notation

$$\tau = \rho_3 = (v_L\ b)\delta, \quad \sigma = \rho_4, \qquad G_{\geq 3} = \mathrm{Alt}((\Omega \setminus D_L) \cup \{v_L\}).$$

The suffix $G_{\geq 5}$ fixes v_L, b; transitivity of this critical tail forces $x = b^\sigma \neq b$. The actual pure-head overlap proof gives $\mathrm{Alt}(W_L) \leq G_{\leq 4}$, where $W_L = D_L \cup \{b, x\}$. It also forces $\tau x = x$: otherwise a second overlap supplies a pure A_4 inside $G_{\leq 4} \cap G_{\geq 3} = \langle \tau, \sigma \rangle$, a dihedral group, which cannot contain noncommuting three-cycles. Hence $\sigma = (b\ x)\epsilon$, where δ, ϵ are odd involutions fixing W_L pointwise. The same actual intersection contains a pure three-cycle on $\{v_L, b, x\}$. Writing $m = |\delta\epsilon|$, the disjoint core/exterior product of $\tau\sigma$ shows that $3 \nmid m$: a pure core rotation requires a multiple of m not divisible by 3. If $m \leq 2$, then δ, ϵ commute, so δ centralizes $\tau, \sigma, G_{\geq 5}$ and fixes v_L, contrary to transitivity of the critical tail and oddness. Thus $m \geq 4$. In particular $J_L = \langle \delta, \epsilon \rangle$ is nonabelian.

Both involutions also fix D_R pointwise. On at most five points an odd involution is a transposition, and two transpositions have product order at most three. Their exterior support therefore needs at least six points. If $n = 18$, only $n - 14 = 4$ points are available outside $W_L \cup D_R$, an immediate contradiction. If $n = 20$, the right crossing $\rho_{r-4} = \rho_6$ commutes with all of $G_{\leq 4}$ and hence fixes W_L pointwise. It sends v_R to a point outside $D_R \cup W_L$ which is fixed by τ and σ, because both fix v_R. That point is also fixed by δ, ϵ, leaving at most five points for their nontrivial action, again impossible.

Finally let $n = 22$, so $r = 11$. The right-hand version supplies W_R of size eight and a nonabelian group J_R generated by two odd exterior involutions. The entire prefixes $G_{\leq 4}$ and $G_{\geq 6}$ commute, so W_L, W_R are disjoint, and both exterior groups act on the remaining six-point set V. Their generators commute, since the original marks commute and the removed core transpositions have disjoint supports.

On six points, an odd involution has either one or three transpositions. Two single transpositions have product order at most three. Two fixed-point-free involutions also have product order at most three, as their union is a disjoint union of alternating even cycles of length at most

six. Thus one of δ, ϵ is a transposition and the other a perfect matching. If the former is an edge of the latter, their product has order at most two. Otherwise their joint orbits have sizes four and two, with the standard D_8 action on four points and C_2 on two points. Its centralizer in S_V lies in $C_2 \times C_2$: on the four-point orbit, centralizing a four-cycle and a reflection leaves only its central involution; the two orbit sizes prevent an exchange. This centralizer is abelian, contradicting the nonabelian J_R centralizing J_L. All possibilities $18 \leq n < 24$ are excluded. $\square$

Exact source route and remaining work. This refines the exceptional continuation of source Propositions 5.28 and 5.32 (physical pages 36–38 and 44–45). The full supported-side hypotheses remain explicit. The D_{10} alternative creates an actual opposite A_5 head, not an automatically unshared next crossing. The resulting natural middle and its anchors are proved, but its maximal permitted rank is not asserted to classify its endpoint markings. The two-ended case for $n \geq 24$, the singleton alternative and shared-label crossings remain open here. Existing complete small-degree certificates and all full predecessor proofs stay paid; no new finite computation or formalization is a premise. No whole L044, original or question credit is claimed.